DIGITAL LOGIC DESIGN — FINAL PROJECT

4-Bit Up/Down Counter with Asynchronous Reset

Synchronous binary counter implementation in Logisim with directional control

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Summary: Concluding slide inviting Viva Voce evaluation and project file repository verification.

Introduction and Project Objective

Sequential Circuit Concept

A 4-bit binary counter cycles through state sequences (0 to 15) using interconnected flip-flops driven by a central clock signal.

Bi-Directional Functionality

Controlled via a direction pin to smoothly shift between incrementing (0000 → 1111) and decrementing (1111 → 0000) counts.

Asynchronous Reset Purpose

Allows immediate override capability to clear all outputs back to 0000 regardless of current clock pulses.

Note Point: Demonstrates how 4 flip-flops provide bi-directional 0–15 binary counting with instantaneous reset override.

Theory and Logic Circuit Design

T Flip-Flop & Logic Gates

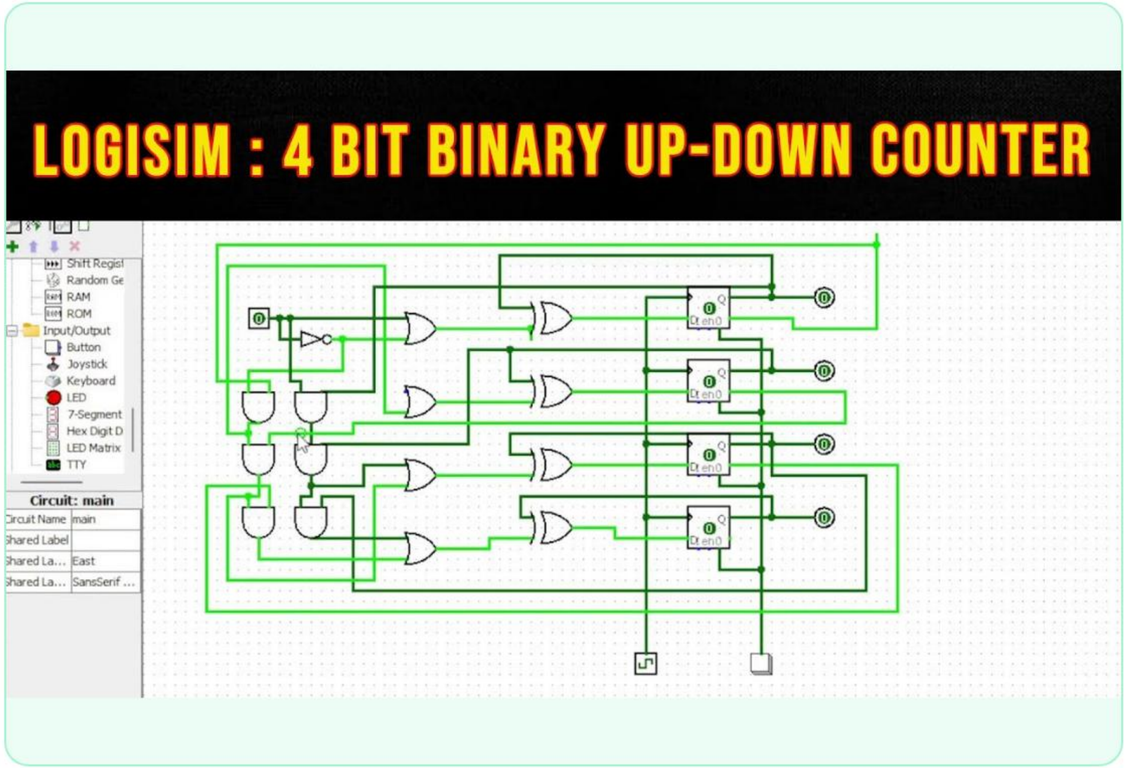
Uses T Flip-Flops paired with AND, OR, and NOT gates to drive state toggles based on previous stage conditions.

Up-Counting Logic (M = 1)

$T_n = Q_{n-1} \cdot \ldots \cdot Q_0 \cdot Up$

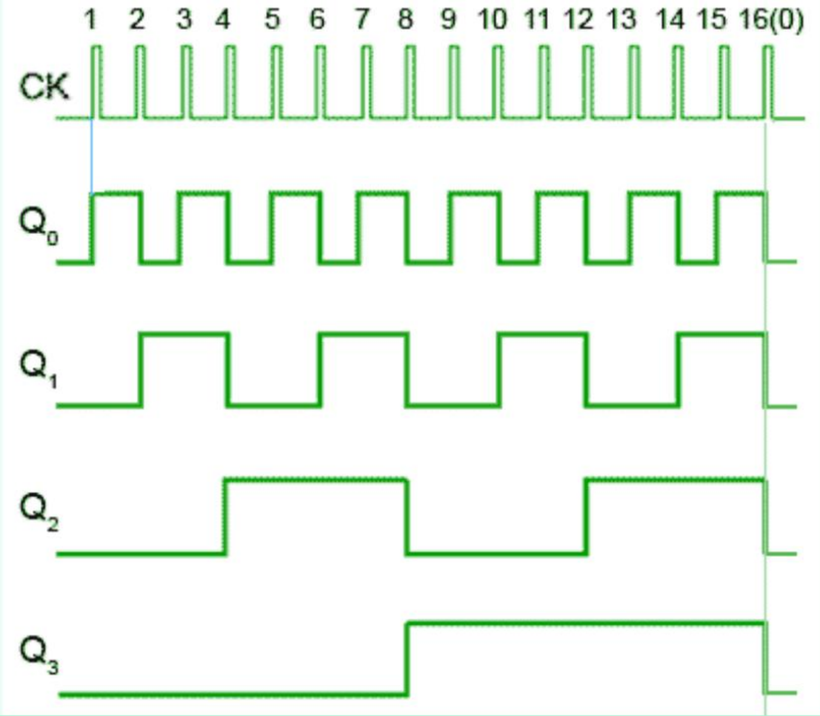
Down-Counting Logic (M = 0)

$T_n = Q'_{n-1} \cdot \ldots \cdot Q'_0 \cdot Down$



Note Point: Steering logic gates control whether Q or Q' signals trigger subsequent T flip-flops to change count direction.

Circuit Design & Working in Logisim



Synchronous Clock Routing

Global clock pulse fires simultaneously across all flip-flops to prevent cumulative delay ripples.

Steering Gate Multiplexing

Combination of AND-OR logic directs normal outputs (Q) or inverted outputs (Q') to subsequent T-inputs.

Asynchronous Override

High reset line directly forces all internal flip-flop clear inputs to low state immediately.

Note Point: Synchronous clocking eliminates propagation delays while Logisim wiring ensures clean real-time state changes.

Simulation Results & Truth Table

Reset	Mode (M)	Clock	Output ($Q_3Q_2Q_1Q_0$)	Working Condition
1	X	X	0000	Asynchronous Reset
0	1	↑	Q + 1 (0000 → 0001)	Up Counting
0	1	↑	1111 → 0000	Overflow Roll-over
0	0	↑	Q - 1 (1111 → 1110)	Down Counting
0	0	↑	0000 → 1111	Underflow Roll-over

Simulation Takeaways

- Toggles cleanly on every active clock edge.
- Mode switch toggles sequence without state corruption.
- Reset instantly zeroes output states across all conditions.

Note Point: Verifies complete truth table compliance, seamless roll-over transitions, and instant reset response.

05 / CONCLUSION & SCOPE

Conclusion and Applications

Parking Systems

Tracks vehicle entry (up) and exit (down) to maintain available slot counts.

Digital Timers

Powers stopwatches and down-counting industrial process controllers.

FIFO Buffer Pointers

Controls stack memory address management in microcontroller registers.

Note Point: Highlights primary practical applications in digital hardware and memory registers along with future scope.

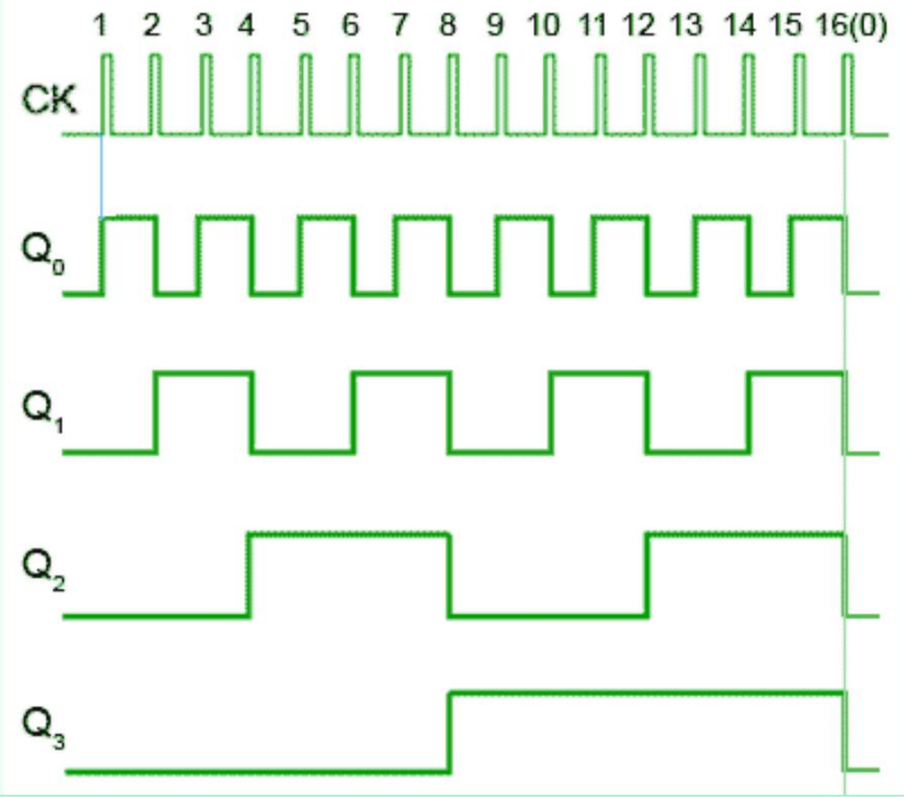


Thank You!

Demonstration completed. We are ready for your Viva Voce questions.

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Circuit Design & Working in Logisim



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Summary: Outlines real-world applications in traffic, timing, and memory systems alongside future FPGA enhancements.

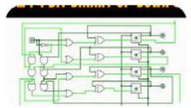


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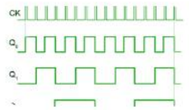
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Image Sources



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